

Data Science
Lab Exercise (Naïve Bayes)

1. Initially perform the following three exercises on the following Links:

<https://www.statology.org/k-fold-cross-validation-in-python/>

<https://www.analyticsvidhya.com/blog/2021/01/a-guide-to-the-naive-bayes-algorithm/>

<https://www.analyticsvidhya.com/blog/2017/09/naive-bayes-explained/>

<https://www.datacamp.com/community/tutorials/naive-bayes-scikit-learn>

2. In the following step, you will be working on building your own Naïve Bayes Classifier.

- a. Create a text file from the following data or you may also use any of the datasets used in the above exercises.
- b. Load data through Pandas libraries
- c. Build your own Naïve Bayes classifier model using steps below
- d. Class: $P(C) = N_c/N$
- e. e.g., $P(\text{No}) = 7/10$,
 $P(\text{Yes}) = 3/10$
- f. For discrete attributes:

$$P(A_i | C_k) = |A_{ik}| / N_{ck}$$

where $|A_{ik}|$ is number of instances having attribute A_i and belongs to class C_k

Examples:

$$P(\text{Status}=\text{Married}|\text{No}) = 4/7$$

$$P(\text{Refund}=\text{Yes}|\text{Yes})=0$$

- g. Normal distribution:

$$P(A_i | c_j) = \frac{1}{\sqrt{2\pi\sigma_{ij}^2}} e^{-\frac{(A_i - \mu_{ij})^2}{2\sigma_{ij}^2}}$$

If Class=No

sample variance = 2975

X1 = {Refund = Yes, Status = Divorced, Income = 90K, Evade = ?}

categorical categorical continuous class

Tid	Refund	Marital Status	Taxable Income	Evade
1	Yes	Single	125K	No
2	No	Married	100K	No
3	No	Single	70K	No
4	Yes	Married	120K	No
5	No	Divorced	95K	Yes
6	No	Married	60K	No
7	Yes	Divorced	220K	No
8	No	Single	85K	Yes
9	No	Married	75K	No
10	No	Single	90K	Yes